DSA Tutorial - 7

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Topic: Evaluation of postfix expression

INPUT :

```
#include <stdio.h>
#include <ctype.h>
#define MAX 100
float stack[MAX];
int top = -1;
void push(float value) {
    stack[++top] = value;
}
float pop() {
    return stack[top--];
}
void evaluateexp(char* exp) {
    float v1, v2;
    int i;

    for (i = 0; exp[i] != '\0'; i++) {
        char ch = exp[i];
        if (isdigit(ch)) {
            push(ch - '0');
        } else {
            v2 = pop();
            v1 = pop();

            switch (ch) {
                case '*':
                    push(v1 * v2);
                    break;
                case '+':
                    push(v1 + v2);
                    break;
                case '-':
                    push(v1 - v2);
```

```
                break;
            case '/':
                push(v1 / v2);
                break;
        }
      }
   }
   printf("Result: %.2f\n", pop());
}

int main() {
   char exp[MAX];
   printf("Enter a Postfix Expression (e.g., 23+4*): ");
   scanf("%s", exp);
   evaluateexp(exp);
   return 0;
}
```

OUTPUT :

